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REVIEW ARTICLE



A review of carbon-pricing studies for developing countries

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ABSTRACT

We review studies that assess hypothetical and actual carbon-pricing schemes in developing countries, focused on policy effectiveness in terms of emissions reduction. The article opens with gathering general reasons for employing this instrument and inspecting if these reasons apply to developing countries. In addition, we consider the implications of unique features of such countries for the design of carbon pricing, notably the choice between carbon taxation and market. Next, results from carbon-pricing studies undertaken for specific low- and middle-income countries are summarized, with an extended appendix providing detailed information per country, organized by main world regions. We find that, since the instrument has seen little application so far, most assessment studies take the form of ex-ante assessment, except for China. The review ends with an assessment of the role of international context and coordination, focusing on carbon markets and climate finance.

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1. Introduction

Compared to the Kyoto Protocol, The Paris Climate Agreement means a more serious involvement of developing countries in reducing greenhouse gas emissions. One motivation for this is that without serious climate policy, emissions growth in coming decades will predominantly come from low- and especially medium-income countries (GCA, 2022), even if the current Paris pledges are implemented through consistent and effective climate policies (King & van den Bergh, 2019; Roelfsema et al., 2020). The share of these countries is already beyond 60% of global carbon emissions (Ritchie, 2023). Hence, it is important to be ambitious in steering development and growth in a low-carbon direction through adequate policies. This is not to deny the historical responsibility and critical role of developed countries in mitigating climate change. They will need to assure significant financial transfers to support for loss and damage, adaptation and mitigation in developing nations. Only then can we hope for achieving implementation of effective climate policies in all countries. This is needed to limit serious carbon leakage as that would hamper the overall, global effectiveness of national climate strategies. Hence, policy stringency should not differentiate too much between developed and developing countries.

There is a large and expanding literature on climate mitigation policy. The present paper focuses its attention on carbon pricing (including taxes and markets) as it is widely regarded as a key element of an effective and affordable climate policy and the best candidate for international harmonization of policies (Aldy & Stavins, 2012; Baranzini et al., 2017; Cramton et al., 2017; Timilsina, 2022). Carbon pricing assures that energy sources and fuels will have a price component

proportional to their carbon content. In effect, through carbon pricing the external cost of carbon emissions over the lifecycle of goods and services is, depending on the price rate, partly or wholly internalized in prices. This creates an incentive for agents to reduce carbon emissions as long as the carbon price exceeds their marginal abatement costs. Unlike other instruments, the non-abated emissions carry a cost that is passed on in derived products and services. This assures that all economic agents will be incentivized to make different choices, namely away from options with high-carbon emissions over their production and life cycle.

The goal of this article is to provide an overview of studies of hypothetical and implemented carbon pricing schemes in developing countries, covering (mainly) ex-ante and some ex-post studies of impacts on carbon emissions and welfare. Our assessment focuses on policy effectiveness in terms of emissions reduction but will also pay some attention to distributional issues and political support. As the framework for evaluating the role of carbon pricing we use the traditional set of criteria, namely effectiveness (of emissions reduction), efficiency (cost or welfare effects) and equity (impacts on income/welfare distribution and poverty), but also pay attention to issues of particular importance to developing countries, such as supra-national harmonization (to allow for stringent policies without harming competitiveness), impact on informal economy, institutional capability and acceptability. In addition, we will discuss how unique features of developing countries – interpreted here as covering both low- and middle-income countries – may affect policy design. Studies reviewed are varied, and cover methods such as price elasticities, econometric-statistical analysis, quasi-experimental and experimental evaluations, and microsimulations using

equilibrium, agent-based or other modelling techniques. Results are organized by world regions.

A relevant question is whether climate policy will affect growth and development of countries, or whether poverty alleviation and climate mitigation are conflicting goals (Barbier, 2014; Hallegatte & Rozenberg, 2017). As suggested by Fankhauser and Jotzo (2018) and Taconet et al. (2020), the costs of climate solutions could slow down the economic catch-up of poor countries. As low-carbon energy sources are more expensive, this could complicate and delay investment in physical infrastructure, negatively affecting development (Jakob & Steckel, 2014). Pricing has a unique advantage of not requiring central planning but leaving some freedom to households and firms, in turn selecting the least-cost or least welfare-harming solutions. This will allow arguably for the smoothest transition and offering the best opportunities for countries' further development.¹ This has also been recognized by a majority of countries participating in the Paris Agreement, reflected by its revised Article 6 during COP26 that aims to establish a global carbon pricing mechanism. The latter seeks to intensify reduction of emissions as well as associated costs, and promotes international linkage between carbon trading schemes (Di Leva & Vaughan, 2021).

To undertake the review, the approach followed in collecting the relevant literature was manual search through references in identified studies, i.e. a snowball approach. We did not opt for a systemic review approach based on keyword search as we felt this was complicated given that we found relevant publications to often not refer in the title and abstract to the low-income or development nature of the country studied. Instead, many studies simply referred to the country name. As a result, our approach has to be judged as having an exploratory nature. Nevertheless, we tried to be complete in coverage of relevant literature mentioned in each study collected, so we feel the snowball effect worked out well. Moreover, various expert reviewers who evaluated the study – including those invited by UNDP (see the acknowledgements) – did not find important omissions. In other words, the review seems quite complete.

The remainder of the paper is organized as follows. Section 2 discusses general insights from the literature about the implementation, advantages, consequences and forms of carbon pricing in developing countries. Section 3 provides an overview of carbon-pricing implementations, plans and studies for specific countries, with an appendix presenting more details organized by main world regions. Section 4 addresses the international context of carbon pricing in developing countries, with particular attention given to international climate finance and carbon markets. A final section 5 draws conclusions.

2. General insights about carbon pricing in developing countries

Carbon pricing is regulatory in nature, in the sense that it changes behaviours or carbon-relevant choices of people and organizations. In other words, its main purpose is to create an appropriate price gap between high- and low-carbon alternatives, not to generate public funds as many other

taxes. The main advantages of using carbon pricing in climate policy have been well documented (Aldy et al., 2010; Aldy & Stavins, 2012; Baranzini et al., 2017; Cramton et al., 2017; Edenhofer et al., 2015; Goulder & Parry, 2008; Metcalf, 2021; Weitzman, 2014, 2017):

1. It is effective because of its systemic nature and particularly if implemented upstream, permeating through all agents (producers, investors, innovators and consumers) and all prices in the economy. Empirical evidence reviewed in Köppl and Schratzenstaller (2022) and van den Bergh and Savin (2021) underpins its effectiveness.
2. It will limit indirect carbon emissions, and hence rebound effects, in complex production chains and lifecycles, which tend to weaken the effectiveness of other strategies and instruments.
3. It accounts for the fact that most consumers, when making purchasing decisions, are more influenced by prices than by environmental concerns.
4. It is cost-effective as it decentralizes policy, thus giving freedom to polluters to select the most affordable ways to reduce emissions, thus also being congruent with a liberal society.
5. It overcomes information limits and associated costs of centralized regulation such as through performance or technical standards. The latter require considerable information from regulators about technical options and costs in all relevant industries.
6. It provides a continuous incentive for adoption and innovation of low-carbon technologies as the carbon price means any remaining emissions have a cost. As opposed, with standards, once they are met, there is no such incentive.
7. While on its own, carbon pricing can be regressive for most developed and progressive for many developing countries (because rich households consume relatively much energy), it becomes more equitable (or even neutral, i.e. undoing the regressive effect) if the revenues of carbon taxes or permit sales in cap-and-trade are (partially) used to compensate inequitable outcomes. Revenues can be especially used to assure poor households are compensated for higher costs of meeting basic needs. Assuring equitable outcomes through revenue use in turn can positively contribute to political feasibility.
8. It is transparent and hence relatively easy to compare, negotiate and harmonize among jurisdictions.

Given the aim of this review, a relevant question is whether these arguments apply equally to low- and middle-income countries. We think most of them do. Argument 1 and 2 are particularly relevant as it will steer the development of these countries, which will go through fundamental changes in coming decades anyway, to a low-carbon economy. Argument 3 is also highly relevant as purchasing power of most households in these countries is limited, and hence minimizing costs (argument 4) for society is of crucial importance. Since governments of developing countries already often struggle with managing the complex economy, argument 5 also carries a lot of weight. Only argument 6 is possibly less relevant to

low/medium-income countries than to high-income countries. Argument 7 is highly relevant as it may help to get public support, and thus contribute to political feasibility of implementing carbon pricing, in poorer countries. Finally, argument 8 is by definition relevant to all countries, including low- and middle-income ones. So altogether, we think the case for carbon pricing in low- and middle-income countries is equally strong as in high-income countries.

In line with the above arguments, around the world over 60 jurisdictions have already implemented carbon taxation or cap-and-trade, among which are several low- and middle-income countries, namely China, Colombia, Kazakhstan, Mexico, South Africa and Ukraine (World Bank, 2023). Others currently consider introducing some form of carbon pricing, notably Brazil, Indonesia, Pakistan, Turkey, Vietnam, Thailand, Senegal, and Côte d'Ivoire (Steckel et al., 2021b). The major success among carbon pricing initiatives is the EU-ETS, covering 30 countries (until recently, before Brexit, even 31). In July 2021, moreover, the European Commission proposed to extend emissions trading to the construction and transport sectors. In the same month, China implemented their carbon market (Nogrady, 2021) – although it should be noted that this is far removed from the textbook proposal (Goulder et al., 2022).

One may wonder if climate policies in low- and medium-income countries – irrespective of whether they focus on carbon pricing or other instruments – should be (very) different from those in developed countries? Given the global commons nature of climate change and to avoid competitiveness, trade and relocation – notably of energy-intensive exporting sectors – and resulting carbon leakage effects, regulator pressure of national policies should not differ much and ideally be identical (Fischer & Fox, 2012; Fowlie, 2009; King & van den Bergh, 2021). Carbon pricing is an instrument of climate policy that allows to compare and harmonize policies, in turn moderating leakage. Perhaps it should not come as a surprise then that few studies question the relevance of carbon pricing in developing countries. An exception is Finon (2019), who suggests carbon pricing might not be appropriate as the main element of a climate policy package in emerging and developing countries because of political economy constraints being greater than in developed countries. He instead suggests the use of non-price instruments and policies such as efficiency standards, regulation, subsidies for clean technologies and public programmes for low-carbon infrastructure. While some of these instruments are potential complements rather than substitutes in terms of mechanism and effect, none of them is systemic and able to sufficiently reduce indirect emissions and contain rebound (van den Bergh et al., 2021). However, although sceptical, Finon concludes that non-pricing instruments serve a useful role in a temporary phase, noting that ‘developing a carbon policy that emphasizes non-price instruments and measures will pave the way to leverage carbon pricing as the main pillar of their future carbon policy in long term’.

A widespread concern about carbon pricing is that it will have inequitable impacts. A relevant study in this regard is Ohlendorf et al. (2021), who examine the distributional impacts of carbon pricing and fossil fuel subsidy through a meta-analysis of 53 empirical studies containing 183 effects

in 39 countries. Progressive distributional outcomes are found for lower income countries and transport sector policies, and when study designs account for indirect effects or demand-side adjustments of consumers. To show more illustrative detail, Dorband et al. (2019) assess expected incidence of a moderate carbon price of US\$30/tCO₂ for different income groups in 87 mostly low- and middle-income countries. They undertake microsimulation using a demand side model with fixed consumption patterns, and a carbon price applied to domestic as well as imported goods and services. The results indicate that for countries with per capita incomes of below USD 15,000 per year (at PPP-adjusted 2011 USD) carbon pricing has, on average, progressive distributional effects. This holds for most countries in Africa and Latin America, as well as the majority of Asian countries considered. Decomposition of impacts shows that outcomes are mainly due to differences among income groups in terms of consumption patterns of energy, and less of food, other goods or services, which seems to confirm that the assumption of fixed consumption patterns was reasonable for this purpose. Their results suggest that climate solutions via carbon pricing are not inconsistent with reducing economic inequality. However, the carbon price may reduce effective income or purchasing power of low-income households, which is estimated to be around 2.5% of income for the considered carbon price.

Additional studies of distributional effects of carbon pricing focus on Asia (Steckel et al., 2021a) and Latin America (Feng et al., 2018). The first examines eight Asian countries, namely Bangladesh, India, Indonesia, Pakistan, Philippines, Thailand, Turkey and Vietnam, combining national household surveys with input-output data. They compare four carbon pricing design options: a globally harmonized carbon price, a national carbon price, and sectoral carbon prices in the power and transport sectors. Findings show considerable variation. Under national carbon pricing distributional outcomes are generally more pronounced within than across income groups, driven by households' energy use patterns. The study by Feng et al. considers subsidy removal and energy taxation by assessing the impact of energy price hikes on income groups using energy-extended input-output analysis. Higher-income groups are found to benefit more from low energy prices than low-income groups when tracing both direct and indirect supply-chain effects of energy price variations. The reason is that such households have more energy-intensive lifestyles and buy more products and services that generate relatively much energy use in their production chain. The authors argue that energy subsidies are a very expensive option to transfer income to poor households, suggesting that in Latin America and the Caribbean transferring \$1 of income through energy subsidies to households in the poorest quintile would cost society then about \$12 (arguably, the largest part of the subsidy ends up with companies). Instead, recycling a small fraction of fiscal revenues from energy subsidy removal or energy taxation through lower income taxes could compensate poor households at a much lower cost. In other words, supporting poor households directly through lower income taxes has a lower social cost than doing it through energy subsidies. This is consistent with the findings of the wider literature on energy subsidies (Couharde & Mouhoud, 2020).

Table 1. Comparing carbon taxation and emissions trading.

Advantages carbon tax	Advantages emissions trading (<i>cap-and-trade or carbon market</i>)	Relevance of arguments for developing countries
Certainty about price as the carbon tax is fixed	Although the price is variable, price stability can be arranged through a price floor & ceiling (safety valve) Endogenous price responding to changes in the economy, preferences and technology The ceiling/cap guarantees effectiveness of emission reduction A carbon markets represents a permanent institution that is stable against political winds and opportunism (especially when international in scope). This explains also why carbon prices are generally higher in carbon markets than with carbon taxation	High as poor people are heavily affected by price changes High as economies of developing countries are becoming gradually more complex and roundabout Medium as otherwise policy will be easily ineffective High as political stability is low in many developing countries
No initial distribution of permits and associated windfall profits for those who have cheap options to reduce emissions		Less relevant as the golden standard is now auctioning or charging a cost for initial permits to avoid windfall profits
Carbon taxes can replace existing energy/fuel taxes so maintain revenues and do not necessarily increase the tax burden		High as tax revenues important for development
Carbon taxation can be part of environmental tax revision (replacing labour by carbon taxes), which can generate a double dividend (employment and environmental benefits). Moreover, this reduces tax evasion and can be designed to compensates inequity effects	Using revenues of permit sales or auction can also be used for compensation of inequity effects	High as revenues are important for a stable government in developing countries
Tax can be gradually increased over time to assure economy has time to adapt	Ceiling to emissions (cap) can be gradually tightened over time to assure economy has time to adapt Easy to internationally extend and harmonize, and thus strengthen (through tightening the ceiling) as competitiveness effects are more limited than with a domestic carbon tax	Medium as a policy shock on the economy is best avoided High as unilateral policy is limited for developing countries

Note: Several of the arguments are discussed and supported with references in the text; for more discussion, see van den Bergh and Botzen (2024).

Alternatively, cash transfers and targeted subsidies for public transportation or food are found to compensate households well. Of course, the political economy of removing energy subsidies is such that most citizens/households (and thus voters) have the perception they benefit from them. This, as well as the fact that many people in developing countries lack an official income, complicates substituting energy subsidies for everyone by lower income taxes for poor households.

Highly relevant to developing countries that depend on international trade or tourism is how carbon pricing will affect emissions as well as the cost of international passenger and freight transport. A few studies have devoted attention to this. They find that carbon pricing is likely to be effective in reducing emissions from international transport. Using changes in bunker prices for maritime trade, which depend on fuel costs, Mundaca et al. (2021) derive relationships of carbon pricing with international product trade and associated emissions. They find that introducing a carbon tax of \$40 would reduce emissions from maritime trade by 7.65%. In addition, their study indicates that a gradual tax increase would allow the International Maritime Organization to reach its emission targets by 2050. A study by Rojon et al. (2021) suggests that while a carbon price will have a limited impact on total maritime transport costs, the average country, Small Island Developing States and Least Developed Countries are likely to be negatively affected. Finally, Zhang and Zhang (2020) examine the impacts of the ETS on tourism-related CO₂ emissions and carbon intensity using a difference-in-differences model. They find that the ETS reduced such

emissions as well as the carbon intensity of tourism and suggest that these effects will strengthen over time.

A unique advantage of carbon pricing, notably taxes, for developing countries is illustrated by Liu (2013). He argues that carbon taxes are more difficult to evade than income taxes as they apply to observable and thus easily traceable fossil fuels while income can be more easily hidden from tax offices. As a result, shifting taxes from the latter to the former – i.e. ‘environmental tax revision’ – can help to reduce tax evasion. Empirical analysis indicates this may decrease social welfare costs of tax evasion up to 28% in the United States, 89% in China, and 97% in India. The latter figures illustrate well that this feature of carbon pricing is particularly relevant to developing countries, also given that such countries tend to have relatively high tax avoidance as a percentage of GDP (Cobham & Janský, 2018). Moreover, any revenue raised from a carbon tax would overwhelmingly come from richer households in these countries, so be equitable within the country. Advani et al. (2021) illustrate this with data for Ethiopia and Ghana, adding that even if carbon pricing is in principle progressive in such countries, this does not preclude that certain poor households might turn out worse off. Hence, compensation is needed, which could be financed by the carbon-pricing revenues. Another risk of carbon pricing in developing countries is that it may encourage the – already widespread – use of firewood, as this can be informally obtained and therefore is insensitive to a price on carbon. Levying a carbon price on cooking fuels might thus push people back into using charcoal or firewood, with negative

consequences not only for climate mitigation but also for human health and gender equality – given that women play a crucial role in collecting firewood. In view also of goals of poverty alleviation, it is not surprising then that many developing countries continue subsidizing fossil fuels (Greve & Lay, 2023; Rose et al., 2022).

The context of certain medium- and low-income countries invites for specific climate strategies. For instance, Brazil and Indonesia possess enormous areas of tropical rainforest, which suggest the importance of specific climate strategies, such as ‘Reducing emissions from deforestation and forest degradation’ (REDD+). The relevance of this is clear as deforestation may contribute up to 7% of worldwide carbon emissions. In view of this, Barbier et al. (2020) propose that countries with considerable tropical forest areas adopt what they call ‘a tropical carbon tax’ on fossil fuels, the proceeds of which are earmarked for investment in ‘natural climate solutions’ – meaning ecosystem conservation, restoration and management which contribute to carbon sequestration. Another motivation is that such mitigation options are relatively cheap in a global context and compared to technological solutions like renewable energy – e.g. one-third less costs per ton CO₂e emissions avoided. Costa Rica and Colombia illustrate potential success, while 12 other main countries with rainforests are suggested as having to follow suit. Barbier et al. advice that governments, development banks, financial investors and non-governmental organizations help these countries to implement such a policy in a correct way.

Finally, given that carbon pricing comes in different forms, namely carbon taxation and emissions trading, an inevitable question is which of these is most suitable for developing countries. While both have seen a wide application worldwide, the most successful implementation in terms of current high carbon prices and international harmonization has been the European Emissions Trading System (EU-ETS): it covers 30 countries and over the past three years its price has most of the time exceeded 80€/tCO₂. An empirical review of average carbon prices, accounting for emissions coverage and exemptions, finds that carbon markets have generated higher average carbon prices than taxation (Finch & van den Bergh, 2022). Especially the EU-ETS has turned out to be a stable institution that has been gradually improved in terms of design over various phases, and virtually untouchable by national governments (even with recent high electricity prices in Europe), which has contributed to its stability. Table 1 summarizes arguments in favour of each of the two types of carbon pricing (Baranzini et al., 2017; Goulder & Schein, 2013; UN Tax Committee, 2021, Chapter 2). It also indicates which arguments are especially relevant to the context of developing countries. While each approach has its own advantages and disadvantages, the balance in our view weighs in favour of emissions trading. This contrasts with emission trading generally meeting more resistance.²

3. National studies of carbon pricing in developing countries

Here we review studies of carbon pricing for distinct world regions and countries. Table 2 summarizes the findings. It

shows that carbon pricing receives attention in all world regions and a wide diversity of countries, namely both low- and middle-income ones. There are, however, few ex-post policy assessments for carbon pricing in low- and middle-income countries. The main exception is a quickly growing number of studies of regional carbon-market pilots in China. However, most other studies pertain to hypothetical or planned carbon taxes or markets and use ex-ante assessment approaches based on ad hoc models which complicates comparison among countries, also as the carbon-pricing designs differ considerably. This is even the case for countries where carbon pricing was implemented. Studies assessing socio-economic impacts generally do not examine the actually implemented carbon price. In some cases, this is because preparatory studies experimented with other prices than implemented (e.g. Mexico); in other cases it is due to late implementation (e.g. Colombia), and in again others (e.g. Chile) studies focus on deriving optimal carbon prices.

Regarding the specific implications of carbon pricing in developing countries, one cannot generalize too much as these countries tend to differ in many regards, in terms of energy and carbon intensity of production, import or export dependence on fossil fuels, importance of international trade for the economy, or trade relationships with other countries. To illustrate, Mattoo et al. (2012) consider impacts of achieving climate goals on manufacturing output and exports. To this end they study a scenario of a modest agreement on emissions reductions, i.e. 17% by 2020 relative to 2005 for industrial countries and 17% relative to business-as-usual increase for developing countries. They find that manufacturing output and exports in low-carbon intensity countries, such as Brazil, are less affected, but that high-carbon intensity countries, such as China and India, face considerable decreases in output (3–3.5%) and manufacturing exports (5.5–7%). Evidently, fossil fuel importers will look very differently at climate policies and carbon pricing than fossil fuel exporters, and further differences in attitudes may arise from differences in average income level.

Initially we aimed at deriving emission-reduction elasticities of carbon pricing for distinct developing countries and then comparing these. However, this turned out to be impossible as few studies derive such elasticities and most provide insufficient information to derive them ex post, or present distinct conditional impact estimates, or focus on a carbon price path. Moreover, such elasticities are easily misinterpreted as reflecting an absolute truth; but they are not constant along the abatement cost curve, and hence depend uniquely on initial emission and price values that differ among countries and studies. In other words, they give a partial picture and are incomparable among countries/studies.³ As a result, our assessment here is descriptive and selective, while specific details for the various world regions and associated countries are provided in the appendix.

Most studies relate to national carbon taxes rather than national markets, and in line with this a recent UN report emphasizes the role of taxes (UN Tax Committee, 2021). Limited institutional capability affects the instrumental choice regarding carbon pricing in low- and middle-income countries. For instance, one might argue that an emissions

Table 2. Overview of carbon-pricing impacts per country/region (details for each row are in associated sub-sections of the Appendix).

World region	Country	Carbon pricing implemented (year: nominal/average prices*)	Emissions reduction (versus benchmark or initial emissions level) and associated GDP losses	Other economic impacts	Methods used	Sources and details
Asia	China	2021: N.A./N.A.	4.3% CO ₂ emissions reduction for a carbon price of 117RM (18US\$), up to –24.2%	Cost-effectiveness 34% higher than for conventional cap-and-trade system	Econometrics (mostly difference-in-differences), also CGE and Energy modelling	Appendix A1
	Other: India, Indonesia, Turkey, ASEAN members, Malaysia, Sri Lanka, Middle East and North Africa (MENA)	None	0.8–8.7% of GDP loss until 2030 for 21% emissions reduction (\$70/t CO ₂ tax); 5% reduction without significant loss in GDP (Turkey) 0.1% GDP loss for 3.6% emissions reduction (India) 24% GHG emissions reduction 2050 (Sri Lanka) Optimal carbon price path 33–68US\$/tCO ₂ in 2020 to 101–225 in 2050 (Malaysia)	Production losses in fossil fuel-intensive sectors (Turkey). Income distribution becomes slightly more inequitable, namely rises with 0.06 points until 2040 (India). No macroeconomic effects assessed for the other countries.	CGE and IAM modelling	
Africa	South Africa	2019: 7.40/N.A.	Carbon price path \$5(R48)/tCO ₂ in 2016, increasing to \$12(R120)/tCO ₂ in 2025, would reduce in 2040 emissions in the power sector with 5% and GDP with 0.7%. Emissions tipping point between \$10 and \$20: at \$10 emissions reductions 5% by 2040, at \$20 it is 50%, and \$50 62%, with GDP loss in range 0.7 to 6.1% (by 2040).	Economic impacts (employment, poverty) are largely neutral under appropriate revenue recycling or carbon border adjustments Economic costs more serious above carbon price US\$20, reducing welfare and employment up to 1.2 and 2.6%	CGE modelling	Appendix A2
	Countries connected to the Zambezi river basin: Malawi, Mozambique, and Zambia	No information provided	No quantitative results for emissions provided, despite the paper mentioning a model for ‘Emissions Prediction and Policy Analysis’ and a ‘scenario characterized by successful global mitigation policy that stabilizes atmospheric concentrations to about 560 ppm CO ₂ eq’.	Carbon pricing, generate two sources of benefit: more favourable and less variable economic outcomes; and, surprisingly, lower fossil fuel prices. These gains exceed mitigation costs	Integrated assessment modelling	
Latin America	Mexico	2014: 2.11/1.01	A rising carbon tax from 2030 to 2050 of 1500 MX\$2015 (US\$100) to 10,500 MX\$2015 (US\$700) results in national goals of 40 and 75% emissions reduction from energy consumption. With no compensation, this reduces GDP by more than 4% after 2040. With redistribution of tax revenues, a double dividend in terms of GDP and emissions reduction results.	Investment and employment drop by more than 4% by 2040. If the policy is unilateral, exports decrease by 8% by 2040.	CGE modelling	Appendix A3
	Brazil	None	Carbon taxes of R\$25–50 per ton of CO ₂ e reduce emissions with approximately 3–6% (7.5–10.5%), while GDP increases slightly with 0.29% or drops up to 3% (–2.1 to –5.4%).	Unemployment reduces with –1.1 to –3.8% (–3.9–6.7%). Inequality falls or increases depending on revenue recycling.	Input-output modelling	
	Colombia	2016: 4.40/1.69	ETS with complete sectoral coverage to achieve 20% emissions reduction versus BAU, resulting in 37 and 41.4 US\$/tCO ₂ e. This reduces	Production in the regulated sectors falls. Higher-income households are the most affected and middle-class ones the least	CGE and IAM modelling	

(Continued)

Table 2. Continued.

World region	Country	Carbon pricing implemented (year: nominal/average prices*)	Emissions reduction (versus benchmark or initial emissions level) and associated GDP losses	Other economic impacts	Methods used	Sources and details
	<i>Chile</i>	2014: 4.40/1.35	annual GDP growth with 0.8%. A 20% emission reduction causes a loss in GDP of 2% (and 2.3% if the tax is extended to consumers), and requires an effective tax rate of about 26 US\$ per ton of CO ₂ e 20 US\$/tCO ₂ e would decrease annual GDP growth by 0.13%	Most affected sectors are oil refinery, transport, and electricity, showing contraction of 7–9%. Electricity prices for consumers increase by 8%.	CGE modelling	
<i>Eastern Europe</i>	<i>Hungary, Slovakia, Czech Republic, Poland, Estonia</i>	<i>Hungary 2005: 14.96/4.24 Slovakia 2005: 14.96/5.38 Czech Republic 2005: 14.96/6.66 Poland 2005: 14.96/7.95 Estonia 2005: 14.96/8.31</i>	1% increase in energy tax decreases CO ₂ emissions between 0.112 and 0.140% (7 emerging economies, including Czech Republic, Hungary and Poland). ETS participation caused slight improvement in CO ₂ intensity but not reduction in emissions. (Lithuania)	Double dividend from substituting carbon taxes for labour taxes (Ukraine). Firms participating in EU-ETS have retired some inefficient capital stock and made modest additional investments without negative effects on profitability (Lithuania).	Mostly econometrics (panel data), also CGE	Appendix A4

Note: The table summarizes each world region based on the details as reported in the appendix (which contains all references to relevant studies); *Based on Finch and van den Bergh (2022), prices converted to €.

trading system might be more complicated to monitor and control and a carbon tax easier to administer as countries have experiences with energy taxes they can build upon. On the other hand, experiences with carbon markets show that this may not be true, and the reviewed studies do not give a clear indication either. Tougher political-economy issues that get attention in some of the reviewed studies relate to time consistency of policies and pressure by interest groups, which may result in carbon pricing not delivering the promised results (Kalkuhl et al., 2020).

Of course, there are various barriers to the full or actual implementation of carbon prices, even in the countries in the table. One fundamental issue is how a carbon price works in the context of highly regulated energy markets in developing countries, aimed at keeping electricity prices low and not being passed through (Borenstein & Kellogg, 2022). Another is that developing countries' production sectors tend to have high carbon intensities (Ward et al., 2019). What matters too is how carbon pricing will affect the relatively large informal sector in developing countries. A study by Bento et al. (2018) shows that under certain circumstances a carbon price can contribute to reducing the informal sector, namely if the revenues are used to lower specific consumption taxes. In addition, carbon taxes could help raise the low average level of taxation (and thus public revenues for health, education and infrastructure) in developing countries (e.g. Besley & Persson, 2014).

While we paid attention, in line with the reviewed studies, to the political acceptability of carbon pricing in terms of distributional effects, other factors play a role as well (Carattini et al., 2018; Drews & van den Bergh, 2016). For example, one of the few studies of public support for climate policies conducted in various middle-income countries shows that communicating how policies work and who benefits from them can improve citizens' attitudes (Dechezleprêtre et al.,

2022). Other important factors to garner public support include targeted transfers – potentially building on previous poverty-eradication programmes (Cameron & Shah, 2014) – and investing pricing revenues in infrastructure (Dorband et al., 2022; Jakob et al., 2016). Regarding distributional effects, it matters how the relation is between those affected and those with power in political processes. Other political-economy issues that could be given attention to garner more support for carbon pricing are co-benefits and compliance. Co-benefits such as more diverse local renewable energy sources may carry some political weight. It is not evident that compliance is easier with taxes than permits, or reverse. Probably there is not much of a difference as in both cases regulators would have to check the (invisible) emissions. However, if the permits are on inputs like fossil fuels – the ideal approach – then compliance is guaranteed, either way.

Finally, a note is in order about fossil-fuel subsidies, which are in effect negative carbon prices. While many studies have been done on this, there is still debate on the relative importance versus positive carbon pricing and the political barriers against removing these, van den Bergh et al. (2024) argue that subsidies contribute to a relatively small portion of climate change and local externality problems, likely around 1%, and that reform of fossil-fuel subsidies is hampered by various political and social factors, more so than the implementation of carbon pricing (see also Skovgaard & van Asselt, 2019). This may also be the case in developing countries as here fossil-fuel subsidies to consumers are common to support especially poor rural households. A study by Jewell et al. (2018) concludes anyway that subsidy removal would be most effective in high-income oil- and gas-exporting regions – notably 'Russia+' (covering several economies that were part of the Soviet Union, such as Ukraine, Kazakhstan and Azerbaijan), Middle East and North Africa (MENA) and Latin America.

4. The international context of carbon pricing in developing countries

To understand the potential of climate policy in developing countries, one must consider it in an international or even global context, for various reasons (van den Bergh et al., 2020):

- Developing countries are part of an international trading network, which makes them economically sensitive to increases in production costs due to climate policy, both in domestic and export markets.
- Global effectiveness of national climate policies requires limited international carbon leakage due to trade effects and relocation of industries (Ing & Nicolai, 2019; King and van den Bergh, 2021).
- National climate policies often exempt export sectors, which make up a large part of the economy in many developing countries. To avoid this, some degree of international policy harmonization is needed.
- Harmonization of national climate policies will reduce worries about loss of competitive position, in turn allowing such policies to become more stringent over time.
- Additionally, financial support from rich countries may be needed to foster harmonized and ambitious climate policies. Doing this on the basis of automatic revenues from carbon pricing would create a stable foundation for climate finance, shielding it against political changes in funding countries.
- Last but not least, carbon pricing is a wise option in case of considerable heterogeneity among polluters in terms of abatement options and costs, as both national and global costs of emissions reduction will be much lower than with other instruments. Such heterogeneity is high between groups of developing and developed countries, with the latter having generally relatively cheap emissions abatement options compared to the first.

According to the report of the UN Tax Committee (2021), in developing countries carbon pricing can serve the dual purpose of regulation of greenhouse gas emissions and generating revenues for public investment during a green transition, such as for R&D. This explains why it is on the agenda of many developing countries as well as broader initiatives, such as the Addis Ababa Action Agenda of 2015. The latter created a global framework for aligning financial flows and policies with economic, social, and environmental goals, closely linked to the Sustainable Development Goals. Another relevant context was created by the Paris Climate Agreement under the UN Framework Convention on Climate Change. It involved the international community promising to mobilize at least USD100 billion per year for mitigation and adaptation in developing countries (Steckel et al., 2017). However, until now these promises have not been fully met while also confusion remains about the definition and scope of 'climate finance' – in terms of compensation for loss and damage or funding of mitigation or adaptation (Timperley, 2021).

International finance could supplement national pricing of carbon generating public revenues, for investing in infrastructure providing access to water, sanitation, electricity,

telecommunications and transport. External finance is needed as most countries will be unable to generate sufficient revenues through carbon pricing, notably countries in Sub-Saharan Africa (Jakob et al., 2016). Against this background, various authors consider which design of climate finance is needed – e.g. Bowen (2011) does this from a public-finance angle. Financial support to developing countries, however, risks inducing a 'climate finance curse' similar to the 'resource curse', i.e. adverse effects on the economy and its development due to a country relying heavily on natural resource exports (Jakob & Steckel, 2014). This involves mechanisms such as excessive growth of the public sector, negative effects on economic diversity and education, easy resource revenues triggering corruption and nepotism among those in power, and appreciation of the nation's currency hampering (potential) other export sectors (known as the 'Dutch disease'). In this respect, it is worthwhile noting that traditional project-based development financing, by NGOs, is often supplemented by macro-economic and sector-wide assistance by international organizations like IMF and World Bank, conditional upon policy reforms. This is confirmed by a study of Tirpak and Adams (2008) examining trends in development assistance funding for energy from bilateral and multilateral donors during 1997–2005. During this period, aid for energy was 6–10% of all development assistance, with a slight shift away from fossil fuel to lower greenhouse-gas-emitting projects. The authors suggest increasing funding for multilateral development banks, to 'green' private sector funds, and to standardize collection and reporting of data on energy investment.

Most of the previous authors agree that reconciling the global goal of cost-efficient mitigation with national policy priorities, climate finance for carbon pricing could become a central pillar of sustainable development and promote international cooperation to achieve the climate targets laid down in the Paris Agreement. A study by Budolfson et al. (2021) finds that if all countries adopt a sufficiently high uniform global carbon tax and then return the revenues to their citizens on an equal per capita basis, it will be possible to meet a 2°C target while also increasing wellbeing, reducing inequality and alleviating poverty. As shown in their Figure 1, the carbon tax level for this case is for a very long period slightly above 100 (2005) US\$. This study uses a global climate policy model using cost-benefit analysis (NICE – nested inequalities climate-economy model), dividing the world into 12 regions and each region into 5 income groups. They also derive a so-called carbon Laffer curve (bell-shaped), which plots total global revenue versus global decarbonization %, indicating a maximum around 70%. Along with this, the positive effect of the carbon tax through progressive redistribution is initially strong but diminishes once the economy decarbonizes as then revenues for inequity compensation decline. A related study by Leimbach and Giannousakis (2019) quantifies mitigation costs for several major world regions, considering scenarios with and without climate finance. They show that sustainable development as represented by the SSP1 scenario reduces the challenges of burden sharing and makes it easier to achieve equitable climate policies. In contrast, in a scenario with fossil-fuel intensive development (SSP5), mitigation costs across regions and required international transfers to increase

under most burden-sharing regimes. They conclude that the first scenario is politically less risky and therefore more likely. A related study by Soergel et al. (2021) finds that, without progressive redistribution, climate policies would push an additional 50 million people into poverty. However, redistributing the national carbon pricing revenues domestically as an equal-per-capita climate dividend reverses this to a small net reduction of global poverty (6 million people less). An additional international climate finance scheme achieves more substantial poverty reduction globally and particularly in Sub-Saharan Africa.

A similarly ambitious study by Bauer et al. (2020) is motivated by the challenge that international cap-and-trade with a uniform carbon price imposes potentially excessive abatement costs on developing countries.⁴ In view of this, international financial transfers are needed as a complement. These can be designed to achieve equal sharing of effort, i.e. distribution of mitigation costs as a share of income. The authors study which financial transfers would be needed instead if moderate deviations from uniform carbon pricing are allowed for developing countries as this is more realistic, economically precautionary, and politically feasible as a first step in trajectory towards uniform pricing globally. They use the integrated assessment model REMIND-MagPIE to this end, and find that under uniform carbon prices, a present value of international financial transfers of 4.4 trillion US dollars over the next 80 years to 2100 would be required to equalize effort. At the other extreme, achieving equal effort without any compensating international financial support, would require carbon prices in advanced countries to be up to 130 times higher than in developing countries, leading to efficiency losses of 2.6 trillion US dollars. The hybrid solution of moderate deviations from uniform carbon prices strongly reduces financial transfers while suffering from relatively small efficiency losses. Since distinct hybrid solutions (in fact, a continuum of them) are feasible, the authors derive a trade-off curve by identifying the cost for different combinations of regional prices (Figure 2 in Bauer et al., 2020). The inefficiency (and effectiveness) sacrifice of non-uniform carbon prices seems, however, to be underestimated as international carbon leakage is not accounted for.

A study by Wang et al. (2018) explores the scale and benefit of a global carbon market under different effort-sharing principles to achieve the 2°C goal, using an integrated assessment model and an effort-sharing platform. They find that global trading volumes would change from 1.8 to over 12 GtCO₂ per year and largely peak between 2030 and 2040 under different kinds of effort-sharing principles. Correspondingly, annual global finance flows would increase gradually and reach the scale of hundreds of billions United States (US) dollars. Much of these revenues would go to developing countries, helping to accelerate their domestic low-carbon energy transformation under good economic conditions (e.g. job creation). A global carbon market would lower the abatement costs of developing countries during 2011–2050. For example, for countries in the Middle East and Africa, the study finds benefits of 4.6 to 14.8% of GDP, depending on the effort-sharing principle used. The overall global abatement costs would drop by 0.4–2.6%.

A theoretical analysis of a development-compatible refunding system to mitigate climate change is offered by Gersbach and Hummel (2016). They propose industrial countries pay an initial fee to a global fund and each country chooses its own national carbon tax. The global fund is then redistributed among developing and industrial countries in proportion to relative emissions reductions, with refunds being net of national tax revenues. This accomplishes efficient emissions reductions and equity, as developing countries do not have to pay an initial fee and are net receivers of funds, hence are overall net beneficiaries, and abate voluntarily – hence will compare all costs and benefits.

International carbon markets can come about in several ways. One of these is linking existing carbon markets. A study by Gavard et al. (2016) examines this approach, focusing on trade in carbon permits between Europe or the US and China. In line with policy discussions indicating that permits imported by developed from developing countries should be limited, their analysis poses a limit on purchasing carbon from Chinese electricity and energy intensive sectors. Using a global economy-wide model, this limit drives a wedge between the carbon prices in China and Europe/ US. Still, global emission reductions come out higher due to emission trading reducing carbon leakage. Moreover, if China captures the rents associated with limited permit trading, it is possible to find a limit threshold that makes both regions better off relative to independent carbon markets.

In the context of international carbon markets, it is worthwhile mentioning the idea of climate clubs and carbon duties. The idea of an expansive climate club of ambitious nations with a uniform carbon price and protecting its border through a carbon tariff was first analysed by Nordhaus (2015, 2021). This would mean both economic and diplomatic pressure (including at UN COP meetings) to make other countries join the club (van den Bergh et al., 2020). This likely will work best with a carbon market as we suggested (Table 1), also as the emissions trading system of the European Union (kind of a club of EU members itself) could best serve as the pragmatic starting point for a broader club. If the USA and China could get aboard, the rest of the world would be strongly motivated to join as well (Tagliapietra & Wolff, 2021). Not an easy scenario, but the number of negotiation partners in the initial stage would in any case be very limited. It would likely also involve that either carbon prices differ during some transition period between rich and poor nations, or that a global climate fund would compensate the poor nations for high carbon costs.

One could consider the countries participating in the EU-ETS already a kind of a climate club. In fact, it has started the initial testing phase of implementing a border tariff, namely the carbon border adjustment mechanism (CBAM). This will partially protect the EU-ETS against competition from exporters to the EU area and will avoid negative trade and relocation effects. A recent study by Magacho et al. (2024) examines the impacts of the CBAM on low- and middle-income countries. Using trade data and multi-regional input-output matrices, it finds that products coming from Russia, China, Turkey and Ukraine, and to a lesser extent from Balkan (East European) countries, Mozambique,

Zimbabwe, Cameroon, Morocco and Tajikistan, will be the most sensitive to CBAM. The authors suggest strategies to ameliorate adverse socio-economic impacts on these countries, such as transfer of technologies, climate subsidies or concessional lending by the EU to reduce the countries' dependence on highly emitting industries.⁵ The authors further point at the risk for developing countries of richer countries – such as Japan, North America or even China – in the near future also adopting CBAM-like measures.

An interesting view on the challenge of creating a global low-carbon economy is proposed by a study of Bataille et al. (2008). It develops a model covering responses to an exponentially rising carbon price (from US\$10/tCO₂e in 2013 to 100 in 2050) by a slow-growing but highly developed economy (Canada) and a fast-growing developing economy (China). The model (CIMS) is hybrid, technologically explicit (energy efficiency, fuel switching, carbon capture and storage, and accelerated development of renewables), behaviourally realistic and includes macroeconomic feedbacks. The results indicate that a global low-carbon economy is feasible if long-term policy packages are accepted with carbon pricing policy as the core, coupled with supplementary regulations. China could reduce emissions by 50% in 2050, with permits flowing from Canada to China as abatement costs of the former country are lower. This goes a bit against general findings about abatement costs differences which suggest that high-income countries tend to possess the most expensive abatement options (Stern, 2012).

Similarly, focusing on China in an international setting, Huang and Bailis (2015) explore its attitude towards the possibility of global carbon market integration, notably for civil aviation. It is evident that, given China being the largest annual emitter at the country level, its participation would be crucial to the success of any global market-based efforts to reduce GHG emissions. The study argues that China tends to prefer gradually establishing and strengthening international governance mechanisms, as this maintains its sovereignty and policy flexibility. This explains China's resistance to join carbon-trading initiatives of Australia, the EU, the International Civil Aviation Organization (ICAO), and the World Bank.

Finally, a study by Bryan et al. (2010) considers the potential participation of sub-Saharan African (SSA) countries in a global carbon market, notably related to agriculture and forestry. The economic potential for mitigation through agriculture (forestry) in the African region is estimated at 17 (23)% of the total global mitigation potential for the sector. Carbon markets should be expanded to include projects related to agriculture, forestry and other land uses (AFOLU). Difficulties arise, though, as smallholder farmers and herders are not easily reached, so that carbon markets should create clear benefits directed at these actors and domestic policies in SSA be reformed to increase both profitability and sustainable practices for rural communities.

5. Conclusions

Carbon pricing means putting a price on the carbon content of fossil fuels, and so effectively on the emission of greenhouse gases due to combusting the fossil fuels. As a result, energy

and fuel prices will be proportional with their carbon content. This provides the most precise and direct incentive for consumers, producers, investors and innovators to consider direct and indirect emission effects of all their decisions. In turn, this will guarantee both minimal lifecycle emissions and minimal cost of emissions reduction, or maximum reduction for a given cost. In addition, pricing generates revenues for compensating any inequity effects or investing in meeting sustainable development goals. We argued that most of the reasons for carbon pricing apply also, and in some cases very strongly, to developing countries.

Our review finds that carbon pricing is experimented with, or planned for, in various low- and middle-income countries, with a concentration in the latter and with examples in all world regions. Most of the covered studies provide ex-ante analyses of hypothetical carbon taxes. There are very few ex-post policy assessments, apart from China. But the latter all address the same emissions-trading system and thus tend to use overlapping data, mostly covering the same regional pilots in China. Together, the reviewed studies give a rather partial picture, also as findings are not easy to compare between countries or studies. Indeed, few studies present, or allow deriving, general indicators such as emission-reduction elasticities of carbon pricing which would allow comparison. Before drawing final conclusions, one would need to experiment more with higher carbon prices in developing countries and collect evidence on its impacts on emissions and welfare. The more so, as it is highly uncertain how economies characterized by considerable unsatisfied demand (for goods, services and energy) – which invites for the rebound effect – or with a large share of informal activity – which might well grow under regulatory pressure – will respond to a carbon price. For instance, the model studies reviewed generally do not pay attention to either effect. The largely model-based or ex-ante evidence, based on single and multi-country studies, of the effectiveness and efficiency of carbon pricing is considerable, despite low prices thus far.

Regarding empirical or ex-post evidence on the efficacy of carbon pricing, it is good to note that it mainly pertains to developed countries. The evidence for developing countries is meagre, and as indicated above mainly based on ex-ante analysis of hypothetical or planned rather than ex-post analysis of implemented policies. One might argue that insights can be drawn from the broader literature on experiences with energy levies and taxes. However, a disclaimer is that the effect of carbon taxes might be higher due to a so-called 'salience effect'. For example, using a natural experiment for the Canadian province of Alberta, Ahundjanov and Noel (2021) find that incidence is higher for an increase in a carbon levy than a generic excise tax. They suggest that 'benefit-side transparency', notably using the tax to alter behaviour for climate purposes, matters. Other studies find similar effects, such as Rivers and Schaufele (2015) for another Canadian province, British Columbia. It is yet unclear that such findings for a developed country can be transferred to a developing-country context.

In terms of the choice between carbon taxation and markets (cap-and-trade) more generally, a main advantage of a carbon tax is that it can easily build upon existing energy and fuel taxes in developing countries, which means users are accustomed to

them and the relevant administrative structure to implement a carbon tax is already in place. Hence, the best these countries can aim for is at short notice replacing current energy taxes by genuine carbon taxes, and in the longer run strive for incorporation in an international carbon market or cap-and-trade system. The latter has the important advantage of limiting competitiveness losses when such markets include key trade partners in the region. This is highly relevant in view of the large component of imports and exports in modern economies, even of large countries, higher carbon prices – or more generally stringent national climate policies – necessitate international cooperation and ideally harmonization. An illustration of the relevance of carbon markets in this regard is the success of the EU-ETS, the only instrument that has managed to harmonize national climate policies among multiple (in fact 30) countries, resulting in a price fluctuating around 70–80€/tCO₂, since 2022, and even exceeding 100€/tCO₂ in March 2023. Considering a range of additional features, we argued that while carbon taxes and markets have both advantages and disadvantages, in our view these balance in favour of carbon markets (as summarized in Table 1). Of course, it is likely that both types of carbon pricing will be present for the foreseeable future.

Although one might have liked a more synthetic review aggregating individual studies, this can only be done if more ex-post studies using comparable statistical-econometric analysis become available for a wider set of countries (beyond China), which requires that first more countries implement carbon pricing. Nevertheless, we feel the descriptive character of our review here (and in the appendix) and the combination of ex-ante and ex-post studies is informative and will help other researchers – particularly in low- and middle-income countries – to quickly get an overview of, and learn about, methods and results for (hypothetical) carbon pricing in these countries.

Notes

1. Incidentally, it is good to separate between GDP growth and development, as for instance captured by the Human Development Index (HDI), since they can lead to different empirical insights and policy suggestions (van den Bergh & Botzen, 2018; Boroumand et al., 2022). Shifting the narrative of climate policy evaluation from one of GDP growth to a message of improving social welfare, as captured by the HDI, might facilitate political leaders and climate negotiators to publicly commit themselves to ambitious climate goals and policies, including carbon pricing.
2. For another view, see Muhammad (2022) who pleads in favour of carbon taxation in developing countries. He offers an informative table of revenues collection from carbon taxes in 31 countries and provinces between 2016 and 2020.
3. Elasticity estimates obtained by two multi-country studies are as follows: a recent analysis of 142 countries over two decades by Best et al. (2020) finds that a €1 price per ton CO₂ reduces annual emission growth by 0.3%; and for a more limited sample of OECD countries Sen and Vollebergh (2018) finds a higher value of 0.73%.
4. According to Dietz et al. (2018), a range of models suggest that the median global carbon price under 1.5°C scenarios is more than US \$100/tCO₂ in 2020. They note that carbon pricing is needed to reduce mitigation costs substantially compared with ramping up the current patchwork of regulatory instruments. In addition, innovation policies and mitigation finance to developing countries are required.

5. In fact, the EU is giving attention to this. See EU (2024).
6. Some refer to it as an ‘absolute or mass-based emissions cap’, to distinguish it from China’s rate-based approach.
7. In addition, 40 large Indian businesses with a total revenue exceeding \$7 trillion have priced or are planning to price carbon. They include companies from cement and oil industries. <https://www.climatescorecard.org/2020/03/indian-businesses-take-the-lead-on-carbon-pricing/>. For recent arguments in favour of carbon pricing in India see these two essays: <https://www.orfonline.org/expert-speak/pricing-carbon-trade-offs-opportunities-india/> and <https://cadmus.eui.eu/handle/1814/69676>.
8. For technical details, see <https://www.sars.gov.za/customs-and-excise/excise/environmental-levy-products/carbon-tax/>. The tax had a rate of R120/t CO₂ emissions and since January 2020 R127/t CO₂. According to the World Bank Carbon Pricing Dashboard: ‘South Africa’s 2021 carbon tax rate is R134/tCO₂e (US\$7/tCO₂e) and will increase until 2022 by the amount of consumer price inflation plus 2% annually. After 2022, only inflationary adjustments are envisioned.’
9. Other relevant general info about carbon markets in Africa includes: <https://energycapitalpower.com/african-carbon-markets-challenges-and-opportunities>, <https://gga.org/can-carbon-trading-work-in-africa>, <https://www.uclga.org/news-caro/carbon-pricing-in-africa-opportunities-for-action-at-the-territorial-level>, Bryan, E. et al. 2008. Global Carbon Markets: Are There Opportunities for Sub-Saharan Africa? IFPRI Discussion Paper 00832, Konrad-Adenauer-Stiftung (2020). Carbon Pricing in Sub-Saharan Africa. February 2020. <https://www.kas.de/en/web/climate-energy-africa/home>, and Carbon pricing approaches in Eastern and Southern Africa, African Development Bank Group, <https://www.afdb.org/fileadmin/uploads/afdb/Documents/Generic-Documents/Carbon%20Market%20Quick%20Facts%2020ACF%202012.pdf>.
10. <https://www.climatescorecard.org/2020/03/mexicos-well-established-carbon-tax-and-pilot-emissions-trading-system-with-california-and-quebec/>.

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Appendix. Studies of carbon-pricing per region/country

Note: if a region or country is missing, it means we did not find any relevant study.

A.1. Asia

China

In mid-2021, actual implementation of the long-awaited carbon market in China was realized. First designed in 2011, already in 2013, the Chinese government initiated seven pilot ETS programmes in the regions of Beijing, Tianjin, Shanghai, Chongqing, Shenzhen, Guangdong, and Hubei. Not only did this allow China to learn from pilot experience and improve its final market design, but it also provided an empirical basis to undertake academic studies. The features of China's carbon market are described in detail by Stoerk et al. (2019), and compared with other systems in operation, by Narassimhan et al. (2018), and formally-numerically analysed in a recent study by Goulder et al. (2022). It concerns a nationwide CO₂ emissions trading system which is unconventional, i.e. deviates from the proposed textbook system as well as from other carbon markets in the world. The first phase of the system limits itself to China's electricity power sector, covering more than 2000 coal-fired power plants and accounting for over 40% of China's total CO₂ emissions. Electricity prices historically had been set by the government, but continual reform since 1985 has resulted in a third of electricity output now being sold at competitive market prices. The system employs a tradable performance standard (TPS), an instrument with a rate-based cap, i.e. applying to the ratio of emissions to output. As shown by Fischer (2001), this instrument implicitly subsidizes output while it taxes emissions. The number of emissions allowances granted to a facility in each compliance period is the product of its assigned 'benchmark' (maximum allowable emissions intensity, which differ between emitters for distributional goals) and its level of output. The latter means, as opposed to a conventional cap-and-trade (C&T) system⁶ that an emitter can influence its allowance allocation through its choice of output during the compliance period, in turn affecting firms' production decisions, emission levels and indirectly cost-effectiveness and distributional impacts. Implicitly the approach subsidizes electricity output, as it effectively avoids using output-reduction as a mechanism channel for emissions reducing. This encourages electricity power plants with relatively low emissions-output ratios to expand output, which is more compatible with economic growth than C&T.

The mentioned study by Goulder et al. develops a simplified theoretical and a more disaggregated empirical partial-equilibrium model to compare numerical simulations of the TPS's impacts with those of a C&T system with similar coverage and achieving the same national emissions reduction. The central estimate of achieved emissions reduction equals 4.3% of initial emissions (equivalent with 116.8 million tons CO₂) for an allowance price of 117 RMB (±18US\$). Cost-effectiveness is found to be 34% higher than when China would have used a conventional C&T system of emissions permits. The advantage of the system is that

it serves regional distributional objectives, an essential part of centralized Chinese governance. In addition, electricity producers bear a larger share of the overall economic burden under the TPS than under C&T and consumers a smaller share. Whether this is good for the economy and development, and consumers in the long run, is questionable. Finally, the approach results in less intersectoral leakage (to sectors not part of the TPS system) as well as international leakage as electricity prices do not go up as much as under C&T. This might be useful for other countries who consider implementing a carbon market with partial coverage, i.e. not all sectors included. In other words, the shortcomings of the TPS system to the ideal C&T become an advantage if not ideally implemented, i.e. with incomplete coverage. The overall conclusion of the study by Goulder is that 'China's nationwide CO₂ emissions trading system, a tradable performance standard, has the potential to make a very substantial contribution to the world's efforts to confront global climate change'. This is an impressive study which clarifies TPS, its diverse impacts, and its differences with C&T. The results may help both Chinese planners to improve the system and other countries to decide which type of carbon market to implement, given political and economic features.

The innovative and comprehensiveness of Goulder et al.'s study renders obsolete most of the earlier research, based on data generated by China's seven pilot ETSs launched from 2013 on. Nevertheless, to give a flavour of that, a few studies are worth mentioning. Zhou and Li (2019) provide more background to China's carbon market, with informative discussions of different types of carbon markets and of carbon trading pilots in China. Generally, these studies find that China's emission trading system (ETS) has been effective in reducing CO₂ emissions. Based on panel data of the two-digit industry at province level from 2005 to 2015, Hu et al. (2020) adopt a difference-in-differences (DID) econometric model to examine the effects of China's ETS on energy conservation and emission reduction. It finds that energy consumption of the regulated industries in pilot areas decreased by 22.8% and CO₂ emissions by 15.5%, compared to those in nonpilot areas. These effects are mainly driven by improved technical efficiency and changed industrial structure. ETS performs better in areas with high levels of environmental enforcement and marketization. Numerous studies examine the impact of China's ETS on carbon emissions and economic growth. Using the common DID method and Data envelopment analysis (DEA), Zhang et al. (2020) find for the period 2014–2016 an economic dividend of 13.6% higher gross industrial output value and a reduction of 24.2% in industrial CO₂ emissions in the seven pilot regions. Similarly, a study by Dong et al. (2019) examines the two dividends – economic and environmental (what they refer to as the Porter hypothesis) as a consequence of the pilot ETS of China since 2013. Also analysing data using the DID method they find no (short-term) Porter effect. However, using a nonparametric mathematical programming technology model they find the Porter effect as a potential long-term outcome if the ETS is applied to the whole of China.

Since China and its economy are so large, one can find studies for China on topics that when studied for other parts of the world usually employ data from multiple countries. An example is a study by Gao et al. (2020) on how China's ETS affects carbon emissions and carbon leakage. The study covers 28 industries in 30 provinces during 2005–2015, using provincial environmentally extended input–output tables and difference-in-differences econometric models. Findings are that ETS contributes to emissions mitigation, has greater effect on the mitigation of production-based emissions than consumption-based emissions, encourages outsourcing of emissions from pilot areas to non-pilot areas (carbon leakage). Other questions have been addressed in the multitude of studies addressing China's ETS. Zhang et al. (2018) examine carbon price volatility in China using data on carbon spot prices from November 2013 to May 2017 in the seven carbon pilots. They find that the Chinese carbon market fluctuated severely at the beginning of carbon trading but has stabilized subsequently. They also note that volatility reacts more severely to good than bad news. They provide lessons for other developing countries. Finally, Weng et al. (2018) argue that to guarantee that China's carbon price signal is strong enough to support the achievement of China's climate change targets given uncertainties about long term economic growth, its carbon market needs a carbon price floor. The authors simulate the China-in-Global Energy Model (C-GEM) and find that a price

floor path of \$4 before 2020, \$8 between 2021 and 2025 and \$12 between 2026 and 2030 (2011 constant price) is able to guarantee China's climate pledges with a 90% chance.

Other Asian countries

Although India is a large emitter, it only recently is considering implementation of pricing. In July 2024, the Indian government adopted a detailed plan for a Carbon Credit Trading Scheme (CCTS), as explained in a draft report of The Bureau of Energy Efficiency (BEE, 2023). The scheme intends to cover sectors like aluminium, cement, fertilizer, iron and steel, pulp and paper, textile, thermal power plant, petroleum refinery, and hotels. Given the recent nature of this plan, few studies have examined carbon pricing and its potential impacts for India. Ojha et al. (2020) argue that India's disinterest in a carbon tax is because of supposed detrimental effects on economic growth and the income distribution. They suggest that this conflict can be partially overcome through a double dividend from a carbon tax, which depends upon how the tax revenue is recycled. To study this, the authors study a carbon tax for 'inclusive green growth' by exploring use of tax revenue for investment to build capacity in either all sectors or exclusively in clean energy sectors, and through transfers to poor households. Their analysis is based on a recursively dynamic CGE model for India with disaggregated energy sectors and an endogenous income distribution module. The findings are that a carbon tax, no matter how it is recycled, cannot promote inclusive green growth, in spite of some double dividend effect; investment in clean energy sectors is beneficial only in terms of climate change mitigation, but not in terms of GDP or income distribution; fears of major GDP losses due to carbon taxes are unwarranted: GDP losses are less than 0.1% for emissions reduction of a little more than 3.6%. Employment effects cannot be estimated as the model assumes full employment. Income distribution becomes slightly more inequitable, namely rises with 0.06 points until 2040.

Indonesia, is one of the largest carbon emitters among developing countries. Yusuf and Resosudarmo (2015) analyse the distributional impact of a carbon tax in Indonesia using a computable general equilibrium model with highly disaggregated household groups. They find that a carbon tax is not necessarily regressive, affecting relatively much urban (versus rural) and higher-income households. The study also shows that revenue-recycling through a uniform reduction in the commodity tax rate may reduce the adverse aggregate output effect, whereas uniform lump-sum transfers may enhance progressivity. See further Steckel et al. (2021a) who study distributional aspects of carbon pricing in eight key countries in developing Asia, namely Bangladesh, India, Indonesia, Pakistan, Philippines, Thailand, Turkey and Vietnam, finding also regressive outcomes.

Another large emitter is Turkey. Employing a computable general equilibrium model, Karapinar et al. (2019) examine how carbon pricing, either through a carbon tax or a national ETS would affect Turkey's economy and emissions. The modelling reflects the three main policy options for the government by 2030, in the context of Turkey's Intended Nationally Determined Contribution (INDC) under the Paris Agreement. This comes down to reducing GHG emissions by up to 21% from its business-as-usual (BAU) scenario in 2030. This is aimed to be achieved through: (i) improving the productivity of renewable energy by 1% per annum, (ii) introducing a tax of 15% per ton of CO₂ originating from carbon-intensive sectors, and (iii) introducing an ETS with caps on emission permits. The policies reduce the shares of agriculture, industry, and transportation, but not more than 1%, while the service sector benefits most. In addition, it is found that unemployment rates are lower, economic growth is stronger, and households are better off under an ETS than under carbon taxation. Another study by Kat et al. (2018) examined the potential implications of an emission trading system with a CGE model with a disaggregated power sector representation. Using a 4% GDP growth trajectory, higher than IMF estimations but lower than the Turkish government's projections, under an ETS emissions would be 21% lower than under a BAU scenarios, and coal-fired power generation would vanish by 2030 due to the high cost of carbon: namely \$50/t CO₂ market price with nuclear and \$70/t CO₂ without. The economic cost would amount to 0.8–1% of the GDP by 2030. A third study by Acar et al. (2016) examined potential macroeconomic impacts of a carbon tax, finding a 21%

reduction in CO₂ emissions compared to a reference scenario and a tax burden of 4.62% of national income by 2030. A carbon tax levied on energy use would raise production losses in fossil fuel-intensive sectors, amounting to an 8.7% decrease in total national income by 2030. Acar and Yeldan (2016) also examine the impacts of coal subsidies in Turkey, using a general equilibrium analysis. They find that through eliminating coal subsidies, Turkey can reduce its emissions by as much as 5% without a significant loss in its GDP. Finally, Gevrek and Uyduranoglu (2015) undertakes a study of public preferences regarding carbon tax design in Turkey. It finds that people dislike personal costs and prefer a progressive impact, earmarking tax revenues (possibly reflecting distrust in the government), and information provision about carbon tax effectiveness, with earmarking having the largest weight.

A study by Nurdianto and Resosudarmo (2016) analyses the benefits and losses associated with cooperation among ASEAN members (Brunei, Myanmar, Cambodia, Timor-Leste, Indonesia, Laos, Malaysia, the Philippines, Singapore, Thailand and Vietnam) in mitigating their CO₂ emissions, particularly by implementing a uniform carbon tax. To this end a multi-country computable general equilibrium model, the Inter-Regional System of Analysis for ASEAN (IRSA-ASEAN) model is used. The results indicate that the implementation of a uniform carbon tax will effectively reduce carbon emissions in the whole region, but also give rise to GDP contraction and reduction in household income. Indonesia and Malaysia can potentially gain as they have already significant energy subsidies.

Using the well-known PAGE09 integrated assessment model (IAM), Wong et al. (2016) identify an optimal carbon price path for Malaysia. They find an initial carbon tax in 2020 of about US\$68.40/tCO₂ in constant 2005 US\$ prices under the A1B Business-as-Usual (BAU) scenario, rising steadily to around US\$225.14/tCO₂ in 2050. However, this study does not examine emissions reduction by Malaysia itself as it derives the optimal carbon price path from a world model through applying a country-specific (relatively low) social cost of carbon for Malaysia. Under a low-emissions scenario, it would be a lower tax at around US\$33.31/tCO₂ in 2020 rising to US\$100.93/tCO₂ in 2050. For a related broader discussion of the pros and cons of Malaysia implementing a carbon tax, see Wong et al. (2019), while for business managers' opinions in this respect see Devi and Wong (2010). Another quantitative study by Yahoo and Othman (2017) used a computable general equilibrium model with explicit energy-emission linkages to assess the economy-wide and welfare impacts of carbon and energy tax policies in Malaysia. They find that carbon taxation gives the best welfare outcome if its revenues are used for consumer subsidies on household purchases. Carbon taxation is further found to reduce emission more than energy taxation.

A study for Sri Lanka by Fernando and Liyanage (2020) assesses alternative carbon price trajectories based on the fifth Shared Socioeconomic Pathway (SSP5) to achieve the 2°C Paris target for Asia, proposed by three IAM models: AIM/CGA, REMIND-MAGPIE and GCAM. The energy-economic-environmental system of Sri Lanka was modelled using the AIM/Enduse model. It considered both energy supply and demand sectors. The results show that in the BAU scenario the GHG emissions are expected to increase from 19.8MtCO₂e in 2015 to 106.2MtCO₂e in 2050. Out of three carbon tax scenarios (price path starting in 2020 or 2030, and rising up to 200, more than 250 or more than 450 US\$/tCO₂ in 2050). The prices proposed by AIM/CGA come out as the most efficient, reducing final GHG emissions by up to 24% in 2050 as compared to the BAU.

Ideas about carbon pricing in the Middle East are discussed in Al-Attayah Foundation (2020), focusing on Gulf Cooperation Council (GCC) countries. It considers implementing a carbon tax in such countries to be challenging because of economic dominance of energy-intensive industries, considerable energy subsidies in place, inefficient energy systems (but hence also options for cheap emissions reduction), and state monopolies of fossil-fuel industries being common (but hence few players which can be easily monitored). In addition, due to low income and corporate taxes, there is little room for shifting such taxes to carbon taxes (and enjoy a double dividend). However, the report regards the EU's intention to implement a carbon border tax as a threat to Middle Eastern exports of oil, gas and energy-intensive materials and products. By adopting their own carbon pricing, Middle Eastern countries would protect their economies while setting in motion a low-carbon transition. The

Middle East could also connect to the EU-ETS and additionally implement a carbon tax for sectors not covered by it, notably road transport. Joint carbon pricing, whether through a tax or market, would also assure that carbon leakage among GCC countries, relevant as they compete strongly in international fossil fuel markets. Success would further require that subsidies on fossil fuel products are removed, to not have two opposing policies. Pricing revenues could support affected firms to adapt, compensate low-income consumers, and fund renewable energy and energy efficiency projects. In addition, Ezenagu (2021) explores the potential impacts of a carbon tax system for *Middle East and North African (MENA)* economies and evaluates potential legal and logistical barriers to its implementation.

A.2. Africa

South Africa

Not surprisingly, given it has a carbon tax since June 2019,⁸ and there was a long preparation of it, various studies have been devoted to a carbon tax in South Africa, some of them motivating its implementation (Vorster et al., 2011). A broad background to mitigation policies and programmes is offered by Chandrashekeran et al. (2017). They conclude that although South Africa is often sold as an emerging green state, its climate performance is actually rather weak. Nevertheless, as argued by Winkler and Marquard (2010), South Africa represents an interesting case given it is a relatively emissions-intensive developing economy due to its sizeable industrial base. In the initial debate, the choice between emissions trading and carbon taxation was on the table. Goldblatt (2010) offers an overview of the arguments at the time, concluding on the basis of a broad set of policy criteria that taxation works better for a middle-income developing country like South Africa. One important consideration is that South Africa's market concentration of firms in the energy sector may hamper realization of a competitive national carbon market as too few firms will trade permits.

Analysis of (hypothetical) carbon taxation in South Africa starts with the older LTMS (Long Term Mitigation Scenarios) study by Winkler (2007). This involved an assessment of economy-wide impacts through CGE modelling (Kearney, 2008; Pauw, 2007), finding modest impacts on GDP (−2% in 2015), varying employment effects (positive for lower-skilled and negative for higher-skilled workers), and adverse impacts on poverty and welfare which were off-settable by tax revenue recycling via a food subsidy. Several other model studies assessed the macroeconomic impacts of a (hypothetical) carbon tax (Alton et al., 2014; Devarajan et al., 2009; Merven et al., 2014; Van Heerden et al., 2006). They all find that a carbon tax considerably reduces emissions while economic impacts are largely neutral under appropriate revenue recycling.

A more recent study by Alton et al. (2014) assesses impacts of an (at the time) hypothetical carbon tax in South Africa using a dynamic economy-wide model linked to an energy sector model. Its results indicate that a carbon tax rising linearly from 3 to 30 US\$/tCO₂ achieves national emissions reductions targets for 2025. Economic costs are moderate initially but become more serious above taxes of US\$20, reducing welfare and employment up to 1.2% and 0.6% relative to a base scenario. South Africa can limit welfare/employment losses by introducing border carbon adjustments. Distributional outcomes depend on how carbon tax revenues are recycled, resulting in different trade-offs between growth and equity.

The other most recent study by Merven et al. (2014) develop e-SAGE, a CGE model SAGE of the South African economy linked to an energy sector model SATIM enabling electricity supply for the CGE model to be endogenized. SATIM is an inter-temporal bottom-up optimization energy model of South Africa built around the Markal-TIMES platform, using linear or mixed integer programming to solve the least-cost planning problem of meeting projected future energy demand. The approach is used to assess a CO₂ tax starting at \$5(R48)/tCO₂ in 2016 and increasing to \$12(R120)/tCO₂ in 2025, approximating the National Treasury's proposed carbon tax. Results showed that the carbon tax would reduce in 2040 emissions in the power sector with 5% and GDP with 0.7%, and lower job creation with 2.60%. Three additional tax levels of \$10, \$20 and \$50 per ton of CO₂ were examined to identify an emissions tipping

point between \$10 and \$20: while at \$10 emissions reductions is 5% lower by 2040, at \$20 it drops with 50%, and increasing up to \$50 only reduces emissions by another 12%. The highest tax would reduce GDP with 6.1% compared to the lowest tax only 0.7% (all in 2040).

Other countries in Africa

Others have evaluated the potential for carbon markets in Africa more positively. Witness the 'West African Alliance on Carbon Markets and Climate Finance', formed in 2017 by 16 countries with the aim to jointly tackle issues relating to accessing carbon markets and climate finance for their NDC implementation. The countries include *Benin, Burkina Faso, Cape Verde, Gambia, Ghana, Guinea, Guinea-Bissau, Ivory Coast, Liberia, Mali, Mauritania, Niger, Nigeria, Senegal, Sierra Leone, Togo*. Four central themes are (IKI, 2022): carbon markets and pricing, transparency, climate finance and technology transfer. A roundtable was organized in July 2020, the first of its kind in Africa. The short-term practical aim of the Alliance is to create in each member state a national platform for practitioners, to allow consultation and capacity building related to Article 6 of the Paris Agreement about cooperation between countries, notably through linking national carbon trading, to achieve emission reduction targets. However, there is not clear aim for a uniform carbon price in alliance countries, which would be good as it would mean a significant scale (especially if a large economy is Nigeria's is included), in turn offering ample opportunities for costs saving through emissions trading. In addition, this would avoid competitiveness effects and carbon leakage due to considerable overlap between export products from these countries.⁹

Finally, Arndt et al. (2019) use integrated assessment modelling, combining biophysical (climate and ecosystems) and economic modules (including emissions and policy costs), to assess the economic and emissions impact of climate change and global mitigation policies on countries connected to the Zambezi river basin, namely *Malawi, Mozambique, and Zambia*. They find that effective global mitigation policies – unspecified, but arguably carbon pricing, generate two sources of benefit: more favourable and less variable economic outcomes; and, surprisingly, lower fossil fuel prices. These gains exceed mitigation costs, which indicates that developing countries stand to gain from effective global mitigation policies.

A.3. Latin America

Mexico

Mexico's Nationally Determined Contributions (NDCs) set out to reduce greenhouse gas emissions unconditionally with 22% by 2030, and with 36% if there is international support and financing, and 50% by 2050. Since 2014 Mexico has had a carbon tax on fossil fuels at rates of up to MXN 39.80 (US\$ 3.50) per ton of CO₂ (while according to the World Bank's Carbon Pricing Dashboard upper value is MXN65/tCO₂e, or US \$3/tCO₂e, possibly different due to devaluation of MXN). In addition, supported by the governments of California and Quebec, Mexico began a pilot programme ETS in 2019, planned to be fully operational in 2023.¹⁰

Gutierrez González (2022) reviews the international influence of the Mexican Emissions Trading System (ETS). This involves the legal framework of the pilot phase of the ETS in Mexico (under the cap-and-trade principle) covering energy and industry sectors whose emissions exceed 100 MT CO₂. It further aims to learn from the European EU-ETS experiences, notably learning-by-doing and improving gradually in subsequent phases.

Landa Rivera et al. (2016) develop a multi-sector neo-Keynesian computable general equilibrium model to evaluate medium- and long-run transitional effects of carbon taxation on the Mexican economy. The level of the carbon tax is computed to meet national emissions targets of 353 and 175 million of tons of CO₂ in 2030 and 2050, respectively, which requires a carbon tax of 1500 MX\$2015 (US\$100) and 10,500 MX\$2015 (US\$700), respectively. This results in 40 and 75% emissions reduction from energy consumption, respectively. Their study further finds that with no compensation, the taxation reduces emissions by more than 75% by 2050 but at a high cost: GDP drops by more than 4% after 2040. Investment and employment drop by more than 4% by

2040. If the policy is unilateral, exports decrease by 8% by 2040. Under full redistribution of carbon tax revenues, a double dividend in terms of GDP and emissions reduction arises.

Brazil

Brazil does not levy a carbon price nor officially plans to do so. Nevertheless, a study by Grottera et al. (2017) uses a Social Accounting Matrix in 2005 to analyse the impact of implementing a tax per ton of CO₂e on GDP, employment and income distribution in Brazil. The results vary with the tax rate and the use of revenues. The results also show an inversely proportional relationship between the carbon tax level (rates used were R\$25 and R\$50 per ton of CO₂e) and the various impacts, with emissions showing the highest elasticity with respect to the carbon tax, compared to economic impacts. The relationships weaken through recycling of tax revenues.

A study for Brazil uses an input-output model to examine how a carbon tax with lump-sum rebate affects household incomes (Moz-Christofoletti & Pereda, 2021). The study finds for both tax rates USD 40/tCO₂e and USD 80/tCO₂e regressive effects. The authors explain this by low price responsiveness and side effects of carbon pricing on food prices. They therefore call for revenue redistribution to support poor households. Similarly, Freitas et al. (2016) use a price system from a national input-output model for Brazil that incorporates the intensity of GHG emissions, as well as a consumption vector broken down into ten representative households with different income levels. They use this to assess the impact of taxing GHG emissions on the Brazilian economy and different household groups. They find significant emissions reduction, slightly regressive effects and a slightly negative impact on output.

Colombia

Three Latin American countries that have implemented a carbon tax also have been subject to various academic studies in this regard. For Colombia, there are several studies to mention. In its tax reform of 2016, the country introduced a carbon tax of about US\$5/tCO₂e covering fossil fuels responsible for roughly 24% of national emissions, with a focus on transportation and liquid fuels. Calderón et al. (2016) examine implementing carbon taxes and abatement targets for the energy system, comparing baseline and policy scenarios using two integrated assessment partial equilibrium models, TIAM-ECN and GCAM, as well as two general equilibrium models, Phoenix and MEG4C. They note that although the carbon intensity of the energy system in Colombia at the time was low compared to the rest of Latin America, projected rapid economic growth and development may change this. Both instruments are found to result in serious emissions reduction. The electric power sector plays an important role in achieving this through more hydropower, wind turbines, and biomass, coal and natural gas with CO₂ capture and storage. Variation across model results is explained by uncertainty about the prevailing mitigation pathways and by differences in model components, base years and data sources. In addition, a wide range of quantitative details are provided based on the four models which cannot be summarized easily. It seems anyway better to focus here on details of more recent studies after implementation of carbon tax (also as authors overlap).

Using a computable general equilibrium model integrated with household-level analysis based on microsimulation models (MSM), Romero et al. (2018) examine distributive effects of a carbon tax. Higher-income households are the most affected and middle-class ones the least. In the most recent study, Sousa et al. (2020) study implications of a hypothetical Emissions Trading System (ETS) in Colombia, using the general equilibrium model MEG4C. They find that production in the regulated sectors falls and that the sectors substitute away from emission intensive goods. The cap assures 5% emissions reduction in 2015 and reaching 20% emissions reduction in 2030, versus a business-as-usual scenario. The minimum and maximum values reached by the permit price is 37 US\$/tCO₂e in 2030 and 41.4 US\$/tCO₂e in 2022, respectively. From this the authors conclude that carbon taxation at the current low level is far from sufficient to achieve the country's NDC. Using funds from the sale of permits to encourage low-carbon investment, abatements are achieved at a lower cost due to economic transformation effects, and positive co-benefits, notably on sustainable development. Implementation of

an ETS with complete sectoral coverage reduce the annual GDP growth rate with 0.8%.

Argentina and Chile

Although high-income countries according to some classifications (e.g. <http://chartsbin.com/view/24389>) and developing countries according to others (e.g. map 1 in <https://hbs.unctad.org/classifications>), regardless of their development status studies for both countries may be useful also to other Latin American countries. Despite a delicate macroeconomic situation, Argentina implemented its carbon tax in December 2017. Rather than intended to change energy prices considerably, the tax was aimed to increase awareness and reduce other taxes. Gutman (2018) discusses reasons why a carbon tax was a defensible political decision, paying attention to macroeconomic imbalance, structural change in the energy sector, and the trade-off between environmental effectiveness, economic efficiency and political feasibility.

In 2014, Chile implemented a carbon tax of 5 US\$/tCO₂e, restricted to large industrial and power generation sources with thermal power greater than 50MW. To see if the tax is sufficiently high, insights from Espinosa and Fornero (2014) are useful. They use a dynamic stochastic general equilibrium model to study the impact of an optimal tax as approximated by Golosov et al. (2014). Their results indicate find that tax benefits outweigh costs, but welfare gains are marginal (2%). The authors calculate the optimal carbon tax for Chile as \$11.95 per ton of CO₂ in 2021 rising to \$18.1, so considerably higher than the current implementation. These values are, moreover, low compared to Golosov et al. \$59.6, due to Chile being a small open economy (less than 0.3% of global emissions). Another study, by García Benavente (2016), also tries to identify the value of a carbon tax that will allow the achievement of the emission reduction target for Chile, whose government in 2009 announced an official commitment with reducing national greenhouse gas emissions by 20% below a business-as-usual projection by 2020. Using a static computable general equilibrium model of the Chilean economy, a tax on emissions from fossil fuels burned by producers is studied. A 20% emission reduction causes a loss in GDP of 2% (and 2.3% if the tax is extended to consumers) and requires an effective tax rate of about 26 US\$ per ton of CO₂e, which is even higher than the estimate from Espinosa and Fornero. The most affected productive sectors are oil refinery, transport, and electricity, showing a contraction between 7% and 9%. Production of electricity from fossil fuels decreases by 11%, while renewable electricity increases by 43%. Electricity prices for consumers increase by 8%. Mardones and Flores (2017) also suggest that the Chilean tax of US\$5 per ton of CO₂ is wholly ineffective in reducing emissions. They regard the main impact of the carbon tax as raising revenues rather than reducing emissions. A study by Benavides et al. (2015) examines the economy-wide implications of a carbon tax applied on the Chilean electricity sector, combining an energy-sector model with a Dynamic Stochastic General Equilibrium model. As a starting point they take the Chilean carbon tax of 5 US\$/tCO₂e, along with potential future increases (10, 20, 30, 40 and 50 US\$/tCO₂e). An interesting finding of the study is that policy effectiveness depends partly on factors that are out of the control of national policy makers, notably project costs of renewable energy investment and natural gas prices. A carbon tax of 20 US\$/tCO₂e would decrease annual GDP growth by 0.13%. Vera and Sauma (2015) compare the effects of the Chilean carbon tax with energy efficiency measures in the power sector. They find that the tax reduces annual CO₂ emissions with 1% compared with a baseline for 2014–2024, while the marginal cost of power production will increase with 3.4%. In contrast, energy efficiency measures reducing 2% of the power demand of the residential sector could achieve larger reductions in CO₂ emissions, while simultaneously decreasing energy price. A shortcoming of the analysis is that it overlooks rebound effects which are typical of energy efficiency improvements (Brockway et al., 2021). Finally, focusing on agriculture, Mardones and Lipski (2020) assess a carbon tax on CO₂e emissions, with rates varying from \$5 to \$131 USD/tCO₂e. They find that a tax applied only to agriculture makes the sector less competitive, reducing its production level. Real GDP falls slightly up to 0.40% and emissions up to 2.25%. The tax is slightly regressive. Of course, a single-sector tax goes against the basic idea of carbon taxation as a systemic instrument.

Linking emissions trading among Latin American countries

The Paris Agreement has recently underlined the relevance of international cooperation via carbon pricing to reduce fragmentation of national climate policies and strategies and improve both effectiveness (including less leakage) and cost-effectiveness. In this respect, a single Emissions Trading Scheme (ETS) with a uniform carbon price could be an option for *Latin America*. Using a dynamic-recursive computable equilibrium model, Diniz Oliveira et al. (2019) study potential implications of such a supranational ETS, including effects on trade patterns and consequences of revenue recycling. The results show that *Brazil* and *Mexico* would import allowances from the rest of Latin America. Revenue-recycling plays an important role in alleviating excessive policy costs for households. The authors advise that a linked ETS system includes not only energy sectors but other sectors as well, notably land use and forestry-related emissions.

In a more recent related study, the same authors (Diniz Oliveira et al., 2021) tackle this issue by studying a link between an ETS for *Brazil* with a developed region (Europe) and two developing regions (*Latin America* and *China*). They use a global economy-wide model, EPPA6, and cover electricity and energy-intensive sectors of Brazil. The results indicate that linking would achieve a lower cost of emissions reduction, and that Brazil imports emission permits. Linking Brazil with a distinct partner such as Europe results surprisingly in more substantial emissions reductions but also losses in GDP and welfare as both regions have ambitious emission-reduction targets. Linking with China is less costly due to less stringent targets, and with Latin America produces moderate reductions. The authors conclude that there are advantages and disadvantages associated with each proposed trading situation.

A.4. Eastern Europe

Wolde-Rufael and Mulat-Weldemeskel (2021) assess the effectiveness of environmental taxes and stringent environmental policy in reducing CO₂ emissions econometrically in a panel of 7 emerging economies that include the Czech Republic, Greece, Hungary, South Korea, Poland, South Africa, and Turkey. The reason is that for these 7 countries they could find a complete and consistent set of data on both total environmental tax, energy tax, and environmental policy stringency available for the period 1994–2015. They find that a 1% increase in energy tax decreases CO₂ emissions between 0.112% and 0.140%. In addition, they suggest an inverted U-shaped relationship between CO₂ emissions and

environmental policy stringency suggesting that it takes time for environmental policy stringency to be effective.

Ukraine has a very energy intensive economy, reflected in one of the highest levels of carbon intensity (i.e. CO₂ emissions/GDP) in the world. In 2011 the Ukrainian government passed a law to impose a carbon tax on the use of energy commodities, with a current tax level of 0.26 Ukrainian Hryvnia (UAH) per tCO₂ (US\$ 0.02 per tCO₂), the lowest in the world. Using a computable general equilibrium (CGE) model for Ukraine, Frey (2017) assesses the impact of different carbon tax levels on the Ukrainian economy and emissions. Her results indicate that the effects of the current tax are negligible, and that to achieve the set target of 22% emission reduction compared with 2007, a carbon tax should have a rate of UAH 40 (US\$ 3.46 per tCO₂). The study also finds a double dividend from substituting carbon taxes for labour taxes. The author suggests that the results are relevant for other former Soviet Union countries in Eastern Europe and Central Asia as they all struggle with high carbon intensity (see in this respect also a study of carbon taxation and double dividend for Russia, by Orlov et al., 2013). An older study for of carbon taxation for Ukraine by Nekrasenko et al. (2015) offers a broader discussion and suggest using differentiated carbon tax rate in distinct regions to account for diversity of 'natural and economic features'.

In a study for *Lithuania*, Jaraitė and Di Maria (2016) assess the impact of the EU-ETS through econometric analysis of panel data of about 5000 Lithuanian firms between 2003 and 2010. They find that ETS participation did go along with a slight improvement in CO₂ intensity but not a reduction in CO₂ emissions. The latter, however, does not mean no effect, as emissions might have risen (more) under the counterfactual (no participation in EU-ETS). Firms participating are found to have retired part of their less efficient capital stock, to have made modest additional investments from 2010 on, and to not have witnessed negative effects on profitability.

Focusing on the wider *Baltic* region – motivated by completion of a planned electricity interconnection between Lithuania, Sweden and Poland – Bariss et al. (2016) study the pass-through of CO₂ prices to electricity prices. Using multiple regression analysis, they show that after the interconnection, a 1€ change in the price of CO₂ emissions would increase the price of the Baltic power market by 0.67€ (versus that of the Nordic power market by 0.55€). They further find that, in a low-price scenario (7.5€/tCO₂), combined heat and power (CHP) plants are marginal generators emitting 0.25 tons of CO₂ per 1MWh in cogeneration mode, while of the CO₂ price exceeds 20€, the marginal emitter will become oil shale power which emits around 1 ton of CO₂ per 1 MWh of electricity.